

Unit1

1. Define robotics and explain its interdisciplinary nature.
2. Discuss the major historical milestones in the development of robotics.
3. What are the advantages and limitations of using exoskeletons for rehabilitation?
4. How has the field of robotics evolved from mechanical automation to AI-driven systems?
5. Compare and contrast first-generation and fourth-generation robots.
6. Explain the significance of **robotic autonomy levels** and their applications.
7. **Discuss the challenges of applying Asimov's Laws to autonomous systems and AI ethics.**
8. Explain how the DH parameter convention simplifies kinematic calculations.
9. How has industrial automation revolutionized manufacturing through robotics?
10. Discuss the concept of robot rights and whether robots should have legal status.

Unit2

1. **Compare the working principles of DC motors, BLDC motors, and servo motors.**
2. How do LIDAR and RGB-D cameras enable robots to perceive their surroundings?
3. . Define the Jacobian matrix in robotic kinematics and explain its significance.
4. Define robot statics and explain its importance in robotic manipulators.
5. **Discuss how the Jacobian matrix can be used to compute manipulability of a robot.**
6. **Explain the difference between path planning and trajectory planning.**
7. How does the **Jacobian pseudo-inverse** method help in inverse kinematics?
8. Discuss the importance of **minimum jerk trajectories** in robotic motion planning.
9. How does **velocity and acceleration profiling** affect trajectory execution?
10. How does **PWM (Pulse Width Modulation)** control the speed of DC and BLDC motors?

Unit 3

1. **Explain the fundamental principles of feedback control in robotic systems.**
2. How does PID control improve the accuracy of robotic joint motion?
3. Explain the working principle of Pulse Width Modulation (PWM) in motor control.
4. Define motion control in robotics and explain its key components.
5. Explain the concept of **space vector PWM (SVPWM)** in three-phase motor control.
6. What are the challenges of implementing **PWM in high-power robotic systems**?
7. How does **trajectory tracking control** work in robotic manipulators?

8. Explain how **feedforward control** improves motion accuracy in robotic systems.
9. How do robots use feature-based localization in indoor navigation?
10. Compare open-loop vs. closed-loop motion control and their applications.

Unit 4

1. Explain the difference between global and local path planning in robotics.
2. Compare BFS and Dijkstra's algorithm in terms of efficiency and path quality.
3. Discuss the advantages and challenges of **LIDAR-based localization**.
4. Explain how **loop closure detection** improves the accuracy of SLAM algorithms.
5. What are the limitations of **odometry-based localization**, and how can they be mitigated?
6. Discuss how **multi-sensor fusion** improves real-time robot localization and mapping.

7. How does priority queue optimization improve Dijkstra's algorithm?
8. Explain the working principle of Breadth-First Search (BFS) in graph-based path planning.
9. Discuss the impact of grid resolution on A Star pathfinding performance.
10. Explain the fundamental principles of SLAM and its importance in autonomous robotics.